

OS_PRACTICAL

1)

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[50];

    printf("Enter command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(command, command, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child finished.\n");
    }

    return 0;
}
```

```

shreenitha@shreenitha2203:~$ nano p2.c
shreenitha@shreenitha2203:~$ gcc p2.c
shreenitha@shreenitha2203:~$ ./a.out
Enter command: ls
Parent PID: 1365
Child PID: 1368
Child PID: 1368
a.out  a6    dir6    exec3.c  f9      open3.c    pipeex1.c
a1     a7    dir8    f1       linux_task1  ossp       pipeex2.c
a10    a8    dir9    f1.c     linux_task2  p2.c       pipeex2.c
a12    a9    dirz    f2       linux_task3  permission_lab pipeex3.c
a2     abc   ex1     f3       linux_task4  pipe1.c    readwr.c
a2.c   b1    ex1.c   f5       open         pipe2.c    student_portal
a3     b2    exec1.c f6       open1.c      pipe3.c    xyz
a5     dir3  exec2.c f7       open1.c.save pipeex1    z7
Child finished.

```

Linux terminal Commands

Uname :

uname is a Linux command used to display information about the operating system and kernel. It helps identify the Linux system, kernel version, and machine architecture.

```

shreenitha@shreenitha2203:~/ossp$ uname -a
Linux shreenitha2203 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC
2026 x86_64 GNU/Linux

```

Lspu:

lscpu shows information about the **CPU that Linux can see**, such as:

- CPU architecture
- Number of CPUs
- CPU cores
- Threads
- CPU model
- CPU frequency

```

shreenitha@shreenitha2203:~/ossp$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-13420H
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               2
BogoMIPS:               5222.40
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
                        mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl
                        xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma
                        cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c
                        rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
                        ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt
                        clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq
                        rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    288 KiB (6 instances)
L1i:                    192 KiB (6 instances)
L2:                     7.5 MiB (6 instances)
L3:                     12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):     0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:               Mitigation; Enhanced IBRS
Spec rstack overflow:   Not affected
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW s
                        equence; BHI BHI_DIS_S
Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected

```

Lsblk:

```

shreenitha@shreenitha2203:~/ossp$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0      2G  0 disk [SWAP]
sdd   8:48   0      1T  0 disk /mnt/wslg/distro
/
shreenitha@shreenitha2203:~/ossp$ |

```

It shows the **block storage devices** Linux can see, such as:

- Hard disk
- SSD
- Partitions
- Virtual disks

lsblk displays information about block storage devices and their partitions. It shows how Linux identifies and organizes storage resources and provides them to the operating system and applications.

Ps:

ps is used to display information about currently running processes. Each process has a unique Process ID (PID). Linux uses process management to control the execution of programs and allocate system resources such as CPU and memory.

```
shreenitha@shreenitha2203:~/ossp$ ps
  PID TTY          TIME CMD
   319 pts/0        00:00:00 bash
  3017 pts/0        00:00:00 ps
```

Top:

```
shreenitha@shreenitha2203:~/ossp$ top
top - 12:29:02 up 2:59, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7756.6 total, 7035.9 free, 566.7 used, 307.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7189.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24548   15484   11588 S   0.0   0.2   0:01.17 systemd
    2 root        20   0    3180    2212    2072 S   0.0   0.0   0:00.00 init-syste+
    6 root        20   0    3180    2048    1940 S   0.0   0.0   0:00.00 init
   46 root       19  -1   50404   16836   15540 S   0.0   0.2   0:00.46 systemd-jo+
   78 systemd+    20   0   22416   14548   12052 S   0.0   0.2   0:00.11 systemd-re+
   85 root       20   0   35364   12276    9196 S   0.0   0.2   0:03.64 systemd-ud+
  162 root       20   0    2888    1948    1820 S   0.0   0.0   0:00.04 chronyd-st+
  163 root       20   0    4472    3132    2884 S   0.0   0.0   0:00.04 cron
  164 message+   20   0    8792    5360    4644 S   0.0   0.1   0:00.12 dbus-daemon
  173 root       20   0   44096   29720   14576 S   0.0   0.4   0:00.23 networkd-d+
  182 root       20   0   18440    9408    8164 S   0.0   0.1   0:00.15 systemd-lo+
  220 syslog     20   0  220548    5912    4608 S   0.0   0.1   0:00.11 rsyslogd
  224 root       20   0    5492    2744    2556 S   0.0   0.0   0:00.00agetty
  259 _chrony     20   0   23600   11144    7220 S   0.0   0.1   0:00.15 chronyd
  270 root       20   0  123460   32704   18096 S   0.0   0.4   0:00.17 unattended+
  273 _chrony     20   0   12096    2420    1780 S   0.0   0.0   0:00.00 chronyd
  314 root       20   0    3184    1116     980 S   0.0   0.0   0:00.00 SessionLea+
  316 root       20   0    3200    1248    1104 S   0.0   0.0   0:00.92 Relay(319)
  319 shreeni+    20   0    6268    5596    3928 S   0.0   0.1   0:00.53 bash
  322 root       20   0    8644    5612    4772 S   0.0   0.1   0:00.01 login
  382 shreeni+   20   0   22348   13508   10892 S   0.0   0.2   0:00.15 systemd
  384 shreeni+   20   0   22916    4092    2088 S   0.0   0.1   0:00.00 (sd-pam)
  414 shreeni+   20   0    6136    5424    3864 S   0.0   0.1   0:00.04 bash
 1402 polkitd    20   0  306444    8608   7796 S   0.0   0.1   0:00.36 polkitd
 3041 shreeni+   20   0    8180    6044    3928 R   0.0   0.1   0:00.01 top
 3042 root       20   0   35368    6512    3400 S   0.0   0.1   0:00.00 (udev-work+
 3043 root       20   0   35368    6512    3400 S   0.0   0.1   0:00.00 (udev-work+
```

top is a Linux command that continuously displays running processes and their CPU and memory usage. It helps observe how the operating system monitors and manages system resources.

Conclusion

Linux provides an abstraction between hardware and applications. Users and programs can use operating system services without directly controlling the hardware. The commands used in this experiment help us observe how Linux manages CPU, memory, storage and processes.

```

#include<fcntl.h>
#include<stdio.h>
#include<unistd.h>
int main()
{
int fd1,fd2;
char buf[100];
size_t n;
fd1=open("a7",O_RDWR);
fd2=open("z7",O_RDWR);
printf("Files succesfully opened and created\n");
printf("%d\n",fd1);
printf("%d\n",fd2);
n=read(fd1,buf,30);
//printf("%s",buf);
write(fd2,buf,n);
}|

```

```

shreenitha@shreenitha2203:~$ nano open3.c
shreenitha@shreenitha2203:~$ gcc open3.c
shreenitha@shreenitha2203:~$ ./a.out
Files succesfully opened and created
3
4
shreenitha@shreenitha2203:~$ |

```

Strace:

```
shreenitha@shreenitha2203:~/ossp/src$ strace -c ./fork1
Child: PID=1550 PPID=1549
Parent: PID=1549
```

% time	seconds	usecs/call	calls	errors	syscall
22.21	0.000457	457	1		clone
20.41	0.000420	52	8		mmap
10.64	0.000219	73	3		mprotect
6.46	0.000133	44	3		fstat
4.96	0.000102	51	2		pread64
3.98	0.000082	27	3		brk
3.69	0.000076	38	2		openat
3.50	0.000072	72	1		read
3.16	0.000065	32	2		rt_sigprocmask
3.11	0.000064	64	1		write
2.92	0.000060	60	1		munmap
2.77	0.000057	57	1		set_tid_address
2.58	0.000053	26	2		close
2.53	0.000052	52	1		rseq
2.48	0.000051	51	1		getpid
1.36	0.000028	28	1		set_robust_list
1.17	0.000024	24	1		prlimit64
1.12	0.000023	23	1		arch_prctl
0.97	0.000020	20	1		getrandom
0.00	0.000000	0	1	1	access
0.00	0.000000	0	1		execve
100.00	0.002058	54	38	1	total

3)

```
#include<unistd.h>
#include<stdio.h>
int main()
{
    int pid;
    pid=fork();
    if(pid==0){
        printf("Child Process\n");
        printf("Child pid is %d\n",getpid());
    }
    else if(pid>0){
        printf("Parent Process\n");
        printf("Parent pid is %d\n",getpid());
    }
    else if(pid<0
    {
        printf("Failed process\n");
    }
    }|
```

```
shreenitha@shreenitha2203:~$ nano p3.c
shreenitha@shreenitha2203:~$ gcc p3.c
shreenitha@shreenitha2203:~$ ./a.out
Parent Process
Parent pid is 1515
Child Process
Child pid is 1516
shreenitha@shreenitha2203:~$ |
```